kacchiOS — Memory Manager Checklist

Core Memory (must include) & Memory Optimization (bonus)

Name: _____ **ID:** _____

Course: _____ **Date:** _____

1. Core Memory — Stack and Heap Allocation (Must Include)

	Requirement	Implementation (file : function)	How it is verified
✓	**Memory detection**	boot.S: Multiboot flag bit 1, passes EAX/EBX to kmain(); memory.c : mem_init()	Boot banner shows "size from Multiboot"; region end follows QEMU -m (8M / 64M / 128M / 512M)
✓	**Memory layout: kernel, buddy pool, heap/stack region**	memory.c : mem_init() — free memory starts at page-aligned __kernel_end (link.ld)	meminfo prints all three ranges; no overlap with kernel image
✓	**Heap allocation**	memory.c : heap_alloc() — first fit on an address-sorted free list, block splitting, 16-byte header {magic, size, requested, check}	memtest: alignment, upward growth, no overlap, hole reuse, largest-block allocation; memdemo step 1 and 3
✓	**Stack allocation**	memory.c : stack_alloc() — last fit from the highest free block, returns initial stack pointer, 16-byte aligned	memtest: stacks above heap, grow downward, full 4 KB usable; memdemo step 1
✓	**Allocation failure handling**	heap_alloc / stack_alloc return NULL for size 0, overflowing or unsatisfiable requests	memtest: heap_alloc(0), heap_alloc(0xFFFFFFF0), stack_alloc(0)
✓	**README.md documentation**	README.md : section "Memory Manager" (design, layout diagram, API, commands, tests, limitations)	Renamed from Readme.md and extended

2. Memory Optimization — Deallocation, Buddy and Slab (Bonus)

	Requirement	Implementation (file : function)	How it is verified
✓	**Heap deallocation**	memory.c : heap_free() → release_block()	memtest: heap_free OK, freed hole reused
✓	**Stack deallocation**	memory.c : stack_free() → release_block()	memtest: stack memory fully returned and coalesced
✓	**Coalescing of free blocks**	memory.c : release_block() merges with lower and upper neighbour	memtest: free list returns to 1 block; 32-block stress test; memdemo step 4
✓	**Double free / invalid pointer detection**	Magic + checksum headers, header poisoning, overlap check in release_block(); page_info[] flags in buddy; freelist walk in slab	memtest: double free, NULL, misaligned, foreign and interior pointers all return MEM_ERR
✓	**Stack overflow detection**	memory.c : stack_check() — 16-byte canary 0xDEADC0DE at stack bottom	memtest: simulated overflow detected
✓	**Buddy allocator (optimised)**	buddy.c : buddy_alloc(), buddy_free() — orders 0..10 (4 KB..4 MB), split on alloc, merge with buddy (pfn XOR 2^k) on free	memtest: 2^k rounding, exhaust whole pool, merge back to half-pool block; buddyinfo; memdemo step 5
✓	**Slab allocator (optimised)**	slab.c : kmem_cache_create / alloc / free / shrink / destroy — one-page slabs, partial/full/empty lists, O(1) embedded freelist	memtest: dense packing, LIFO reuse, 300 objects over several slabs, empty slabs returned; slabinfo
✓	**kmalloc / kfree front end**	slab.c : kmalloc(), kfree() — size classes 16..1024 B via slab, larger via buddy pages	memtest: kmalloc(24) in 32 B cache, kmalloc(5000) page aligned; memdemo step 6
✓	**Checklist.pdf**	This document	—

3. Testing and Verification

	Requirement	Implementation (file : function)	How it is verified
✓	**Builds without compiler warnings**	make (gcc -m32 -O2 -Wall -Wextra -ffreestanding)	No C warnings, no undefined symbols
✓	**Self tests pass**	memtest.c : mem_selftest() — 62 checks	62 passed, 0 failed at -m 8M, 64M, 128M, 512M
✓	**No memory leaks**	Leak check at the end of mem_selftest()	Free list and buddy pool identical before/after; memtest can be repeated
✓	**Original behaviour kept**	kernel.c : memory_command(), unknown input still echoed	"hello" → "You typed: hello"

4. Files Added / Changed

File	Change
memory.h	New — public memory manager API
memory.c	New — detection, heap & stack allocator, coalescing, stats, demo
buddy.c	New — buddy page allocator
slab.c	New — slab allocator, kmalloc/kfree
memtest.c	New — 62 self tests
mm_internal.h	New — helpers shared by the memory manager
kernel.c	mem_init() call, memory shell commands
boot.S	Request Multiboot memory info, pass magic/info to kmain()
Makefile	New object files added to OBJS
README.md	Renamed from Readme.md; Memory Manager documentation

5. How to Verify

```
make clean && make
make run
kacchiOS> help       # list memory commands
kacchiOS> meminfo    # layout, heap/stack usage, free list
kacchiOS> memdemo    # allocation / free walk-through
kacchiOS> buddyinfo  # buddy free blocks per order
kacchiOS> slabinfo   # slab caches
kacchiOS> memtest    # expect: Result: 62 passed, 0 failed
```